BCA201P — LAB SHEET 03

Foundations of Java Development | BCA III Semester | Session 2026–2027

Java Program Solutions (Questions 1–30)

Each program is written as a standalone Java program. Save each program as Main.java when running it separately.

1. Input and display 5 integers using a 1-D array

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] a = new int[5];
        for (int i = 0; i < 5; i++) a[i] = sc.nextInt();
        for (int x : a) System.out.print(x + " ");
        sc.close();
    }
}
```

2. Input 10 integers and display in reverse order

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] a = new int[10];
        for (int i = 0; i < 10; i++) a[i] = sc.nextInt();
        for (int i = 9; i >= 0; i--) System.out.print(a[i] + " ");
        sc.close();
    }
}
```

3. Find the sum of all elements in a 1-D array

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt(), sum = 0;
        int[] a = new int[n];
        for (int i = 0; i < n; i++) { a[i] = sc.nextInt(); sum += a[i]; }
        System.out.println("Sum = " + sum);
        sc.close();
    }
}
```

4. Find the maximum element in an array

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] a = new int[n];
        for (int i = 0; i < n; i++) a[i] = sc.nextInt();
        int max = a[0];
        for (int x : a) if (x > max) max = x;
        System.out.println("Maximum = " + max);
        sc.close();
    }
}
```

5. Count even and odd numbers in an array

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt(), even = 0, odd = 0;
        int[] a = new int[n];
        for (int i = 0; i < n; i++) a[i] = sc.nextInt();
        for (int x : a) if (x % 2 == 0) even++; else odd++;
        System.out.println("Even = " + even + ", Odd = " + odd);
        sc.close();
    }
}
```

6. Copy all elements from one array to another

```
import java.util.Scanner;
import java.util.Arrays;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] a = new int[n], b = new int[n];
        for (int i = 0; i < n; i++) a[i] = sc.nextInt();
        for (int i = 0; i < n; i++) b[i] = a[i];
        System.out.println(Arrays.toString(b));
        sc.close();
    }
}
```

7. Input and display a 3 x 3 matrix

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[][] a = new int[3][3];
        for (int i = 0; i < 3; i++)
            for (int j = 0; j < 3; j++) a[i][j] = sc.nextInt();
        for (int[] row : a) {
            for (int x : row) System.out.print(x + " ");
            System.out.println();
        }
        sc.close();
    }
}
```

8. Find the sum of all elements in a matrix

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int r = sc.nextInt(), c = sc.nextInt(), sum = 0;
        int[][] a = new int[r][c];
        for (int i = 0; i < r; i++)
            for (int j = 0; j < c; j++) { a[i][j] = sc.nextInt(); sum += a[i][j]; }
        System.out.println("Sum = " + sum);
        sc.close();
    }
}
```

9. Find the sum of each row in a matrix

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int r = sc.nextInt(), c = sc.nextInt();
        int[][] a = new int[r][c];
        for (int i = 0; i < r; i++)
            for (int j = 0; j < c; j++) a[i][j] = sc.nextInt();
        for (int i = 0; i < r; i++) {
            int sum = 0;
            for (int j = 0; j < c; j++) sum += a[i][j];
            System.out.println("Row " + (i + 1) + " sum = " + sum);
        }
        sc.close();
    }
}
```

10. Find the sum of each column in a matrix

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int r = sc.nextInt(), c = sc.nextInt();
        int[][] a = new int[r][c];
        for (int i = 0; i < r; i++)
            for (int j = 0; j < c; j++) a[i][j] = sc.nextInt();
        for (int j = 0; j < c; j++) {
            int sum = 0;
            for (int i = 0; i < r; i++) sum += a[i][j];
            System.out.println("Column " + (j + 1) + " sum = " + sum);
        }
        sc.close();
    }
}
```

11. Transpose a matrix

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int r = sc.nextInt(), c = sc.nextInt();
        int[][] a = new int[r][c];
        for (int i = 0; i < r; i++)
            for (int j = 0; j < c; j++) a[i][j] = sc.nextInt();
        for (int j = 0; j < c; j++) {
            for (int i = 0; i < r; i++) System.out.print(a[i][j] + " ");
            System.out.println();
        }
        sc.close();
    }
}
```

12. Add two matrices of the same size

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int r = sc.nextInt(), c = sc.nextInt();
        int[][] a = new int[r][c], b = new int[r][c];
        for (int i = 0; i < r; i++)
            for (int j = 0; j < c; j++) a[i][j] = sc.nextInt();
        for (int i = 0; i < r; i++)
            for (int j = 0; j < c; j++) b[i][j] = sc.nextInt();
        for (int i = 0; i < r; i++) {
            for (int j = 0; j < c; j++) System.out.print((a[i][j] + b[i][j]) + " ");
            System.out.println();
        }
        sc.close();
    }
}
```

13. Multiply two 3 x 3 matrices

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[][] a = new int[3][3], b = new int[3][3], c = new int[3][3];
        for (int i = 0; i < 3; i++)
            for (int j = 0; j < 3; j++) a[i][j] = sc.nextInt();
        for (int i = 0; i < 3; i++)
            for (int j = 0; j < 3; j++) b[i][j] = sc.nextInt();
        for (int i = 0; i < 3; i++)
            for (int j = 0; j < 3; j++)
                for (int k = 0; k < 3; k++) c[i][j] += a[i][k] * b[k][j];
        for (int[] row : c) {
            for (int x : row) System.out.print(x + " ");
            System.out.println();
        }
        sc.close();
    }
}
```

14. Input and display a 2 x 2 x 2 3-D array

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[][][] a = new int[2][2][2];
        for (int i = 0; i < 2; i++)
            for (int j = 0; j < 2; j++)
                for (int k = 0; k < 2; k++) a[i][j][k] = sc.nextInt();
        for (int i = 0; i < 2; i++) {
            System.out.println("Layer " + (i + 1));
            for (int j = 0; j < 2; j++) {
                for (int k = 0; k < 2; k++) System.out.print(a[i][j][k] + " ");
                System.out.println();
            }
        }
        sc.close();
    }
}
```

15. Find the sum of all elements in a 3-D array

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int x = sc.nextInt(), y = sc.nextInt(), z = sc.nextInt(), sum = 0;
        int[][][] a = new int[x][y][z];
        for (int i = 0; i < x; i++)
            for (int j = 0; j < y; j++)
                for (int k = 0; k < z; k++) { a[i][j][k] = sc.nextInt(); sum += a[i][j][k]; }
        System.out.println("Sum = " + sum);
        sc.close();
    }
}
```

16. Count positive and negative numbers in a 3-D array

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[][][] a = new int[2][2][2];
        int positive = 0, negative = 0;
        for (int i = 0; i < 2; i++)
            for (int j = 0; j < 2; j++)
                for (int k = 0; k < 2; k++) {
                    a[i][j][k] = sc.nextInt();
                    if (a[i][j][k] > 0) positive++;
                    else if (a[i][j][k] < 0) negative++;
                }
        System.out.println("Positive = " + positive + ", Negative = " + negative);
        sc.close();
    }
}
```

17. Find the maximum element in a 3-D array

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[][][] a = new int[2][2][2];
        for (int i = 0; i < 2; i++)
            for (int j = 0; j < 2; j++)
                for (int k = 0; k < 2; k++) a[i][j][k] = sc.nextInt();
        int max = a[0][0][0];
        for (int[][] layer : a)
            for (int[] row : layer)
                for (int x : row) if (x > max) max = x;
        System.out.println("Maximum = " + max);
        sc.close();
    }
}
```

18. Search an element in a 3-D array and display its position

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[][][] a = new int[2][2][2];
        for (int i = 0; i < 2; i++)
            for (int j = 0; j < 2; j++)
                for (int k = 0; k < 2; k++) a[i][j][k] = sc.nextInt();
        int target = sc.nextInt();
        boolean found = false;
        for (int i = 0; i < 2; i++)
            for (int j = 0; j < 2; j++)
                for (int k = 0; k < 2; k++)
                    if (a[i][j][k] == target) {
                        System.out.println("Position: [" + i + "][" + j + "][" + k + "]");
                        found = true;
                    }
        if (!found) System.out.println("Element not found");
        sc.close();
    }
}
```

19. Create and display a jagged array with 3 rows

```
class Main {
    public static void main(String[] args) {
        int[][] a = { {1, 2}, {3, 4, 5}, {6, 7, 8, 9} };
        for (int[] row : a) {
            for (int x : row) System.out.print(x + " ");
            System.out.println();
        }
    }
}
```

20. Store marks where each student has a different number of subjects

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int students = sc.nextInt();
        int[][] marks = new int[students][];
        for (int i = 0; i < students; i++) {
            int subjects = sc.nextInt();
            marks[i] = new int[subjects];
            for (int j = 0; j < subjects; j++) marks[i][j] = sc.nextInt();
        }
        for (int i = 0; i < students; i++) {
            System.out.print("Student " + (i + 1) + ": ");
            for (int mark : marks[i]) System.out.print(mark + " ");
            System.out.println();
        }
        sc.close();
    }
}
```

21. Rotate an array to the left by one position

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] a = new int[n];
        for (int i = 0; i < n; i++) a[i] = sc.nextInt();
        if (n > 0) {
            int first = a[0];
            for (int i = 0; i < n - 1; i++) a[i] = a[i + 1];
            a[n - 1] = first;
        }
        for (int x : a) System.out.print(x + " ");
        sc.close();
    }
}
```

22. Count how many times each element appears in a 1-D array

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] a = new int[n];
        boolean[] visited = new boolean[n];
        for (int i = 0; i < n; i++) a[i] = sc.nextInt();
        for (int i = 0; i < n; i++) {
            if (visited[i]) continue;
            int count = 1;
            for (int j = i + 1; j < n; j++)
                if (a[i] == a[j]) { count++; visited[j] = true; }
            System.out.println(a[i] + " appears " + count + " time(s)");
        }
        sc.close();
    }
}
```

23. Find the second largest distinct element in a 1-D array

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] a = new int[n];
        for (int i = 0; i < n; i++) a[i] = sc.nextInt();
        Integer largest = null, second = null;
        for (int x : a) {
            if (largest == null || x > largest) { second = largest; largest = x; }
            else if (x != largest && (second == null || x > second)) second = x;
        }
        if (second == null) System.out.println("No second largest distinct element");
        else System.out.println("Second largest = " + second);
        sc.close();
    }
}
```

24. Remove duplicate elements without using collections

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] a = new int[n], result = new int[n];
        int size = 0;
        for (int i = 0; i < n; i++) a[i] = sc.nextInt();
        for (int i = 0; i < n; i++) {
            boolean exists = false;
            for (int j = 0; j < size; j++) if (result[j] == a[i]) exists = true;
            if (!exists) result[size++] = a[i];
        }
        for (int i = 0; i < size; i++) System.out.print(result[i] + " ");
        sc.close();
    }
}
```

25. Sum of main and secondary diagonals of a square matrix

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt(), main = 0, secondary = 0;
        int[][] a = new int[n][n];
        for (int i = 0; i < n; i++)
            for (int j = 0; j < n; j++) a[i][j] = sc.nextInt();
        for (int i = 0; i < n; i++) {
            main += a[i][i];
            secondary += a[i][n - 1 - i];
        }
        System.out.println("Main diagonal sum = " + main);
        System.out.println("Secondary diagonal sum = " + secondary);
        sc.close();
    }
}
```

26. Display upper and lower triangular parts of a square matrix

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[][] a = new int[n][n];
        for (int i = 0; i < n; i++)
            for (int j = 0; j < n; j++) a[i][j] = sc.nextInt();
        System.out.println("Upper triangular matrix:");
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) System.out.print((i <= j ? a[i][j] : 0) + " ");
            System.out.println();
        }
        System.out.println("Lower triangular matrix:");
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) System.out.print((i >= j ? a[i][j] : 0) + " ");
            System.out.println();
        }
        sc.close();
    }
}
```

27. Print only the boundary elements of a matrix

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int r = sc.nextInt(), c = sc.nextInt();
        int[][] a = new int[r][c];
        for (int i = 0; i < r; i++)
            for (int j = 0; j < c; j++) a[i][j] = sc.nextInt();
        for (int i = 0; i < r; i++) {
            for (int j = 0; j < c; j++)
                if (i == 0 || i == r - 1 || j == 0 || j == c - 1)
                    System.out.print(a[i][j] + " ");
                else System.out.print("  ");
            System.out.println();
        }
        sc.close();
    }
}
```

28. Search for an element in a 2-D array and show row and column

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int r = sc.nextInt(), c = sc.nextInt();
        int[][] a = new int[r][c];
        for (int i = 0; i < r; i++)
            for (int j = 0; j < c; j++) a[i][j] = sc.nextInt();
        int target = sc.nextInt();
        boolean found = false;
        for (int i = 0; i < r; i++)
            for (int j = 0; j < c; j++)
                if (a[i][j] == target) {
                    System.out.println("Row " + (i + 1) + ", Column " + (j + 1));
                    found = true;
                }
        if (!found) System.out.println("Element not found");
        sc.close();
    }
}
```

29. Sort the elements of each row in a jagged array

```
import java.util.Arrays;
class Main {
    public static void main(String[] args) {
        int[][] a = { {5, 2, 8}, {9, 1}, {7, 4, 6, 3} };
        for (int[] row : a) {
            Arrays.sort(row);
            System.out.println(Arrays.toString(row));
        }
    }
}
```

30. Check whether a matrix is sparse

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int r = sc.nextInt(), c = sc.nextInt(), zeros = 0;
        int[][] a = new int[r][c];
        for (int i = 0; i < r; i++)
            for (int j = 0; j < c; j++) {
                a[i][j] = sc.nextInt();
                if (a[i][j] == 0) zeros++;
            }
        int total = r * c;
        if (zeros > total - zeros) System.out.println("Sparse matrix");
        else System.out.println("Not a sparse matrix");
        sc.close();
    }
}
```